Bedrock

ICU early warning on IBM LinuxONE

System architecture

IBM Z Datathon · RayNCL2023/IBMZ-Datathon @ 9e84a34

Clients

Ward screens
Nurses and doctors in a browser

Live replay of real ICU stays

Demo film `video/`
2-minute Remotion film, 1080p

capture.ts	shots of the live app
data.json	numbers from the API
make_audio	soundtrack, alarm cues

HTTPS

captures

Edge

Bedrock web app `web/`
Next.js on Vinext · Cloudflare Workers

Ward board	cards, table, tiles, bay map
Alarms	acknowledge, escalate, sounds
Patient	risk timeline, drivers, what-if
Handover	Granite SBAR per patient
Shift summary	ward report, last n hours new
Insights	performance, federated
Designs	ward, clinical, night · light and dark

/api/lx proxy forwards an allow-list of endpoints and adds `x-api-key` on the server. The data source can switch to in-browser mock data.

HTTPS + key

IBM LinuxONE

LinuxONE VM · s390x — Docker Compose · 3 cores · nothing leaves the machine

Caddy
HTTPS front door · TLS on CPACF

`:443 TLS` `:80 → 308` `HSTS`

API key store
SQLite · SHA-256 hashed codes

`app.keys create · list · revoke`

verify key

Bedrock API `api/app/`
FastAPI · no published port · every route but /health needs a key, fails closed

`GET /ward` `GET /patient/{id}` `POST /predict` `POST /news2`
`POST /handover/{id}` `POST /shift-summary` `GET /metrics` `GET /privacy`
`GET /fatigue` `GET /benchmark` `GET /health`

IBM Snap ML
NNPA (zdnn_tensors) or CPU trees

Sepsis	6 h
Hypotension	6 h
Acute kidney injury	24 h
Respiratory failure	6 h
In-hospital mortality	stay

IBM Granite 4.0
350M · llama.cpp built with zDNN

Handover	SBAR note per patient
Shift summary	ward, last n hours

Rules build the facts. Numbers not in the facts are flagged; conditions must say "risk".

federated models → model/fed/

Federated training `fed/`

Hospital A 2019 set A — **Hospital B** 2019 set B — **Hospital M** 2012 set — **Coordinator** cyclic XGBoost

The model visits each hospital in turn and gains trees on its own data, with optional Laplace noise (ε dial). Only models, metrics and medians leave a hospital.

PhysioNet data
De-identified ICU stays

2019 A + B	hourly vitals, sepsis label
2012	first 48 h, mortality

ML training `ml/`
Offline, on a laptop

prepare.py	features + every label
train.py	XGBoost 2.1.4, NEWS2-matched

rsync

each hospital mounts only its own data

Live request path · Offline: data, training, capture · new Since the last diagram

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